6. Water treatment is an important process in producing water for drinking. Temporary hard water contains calcium hydrogen carbonate $Ca(HCO_3)_2$ and magnesium hydrogen carbonate $Mg(HCO_3)_2$. Boiling breaks down the hydrogen carbonate to form insoluble carbonates. It is an inexpensive method used to soften temporary hard water.

(a) (i) Complete the equation below to show this process. [2]

$$Ca(HCO_3)_2 \longrightarrow \ldots\ldots\ldots\ldots \ + \ \ldots\ldots\ldots\ldots \ + \ H_2O$$

(ii) Explain why permanent hard water cannot be softened by boiling. [2]

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(b) The addition of sodium carbonate or passing water through an ion exchange column can be used to soften temporary and permanent hard water.

Explain how the following methods soften water.

(i) Addition of sodium carbonate. [2]

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(ii) Passing water through an ion exchange column. [2]

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(c) State **one** disadvantage other than cost associated with an ion exchange column. [1]

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(d) Describe the effects of hard water on the central heating system in a house. [2]

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